Administrator Guide

WebFlowMaster Enterprise Documentation

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🛡️ WebFlowMaster: Enterprise Administrator Guide

> **Governance, Security, and Infrastructure Maintenance**

🎯 Document Objective

This guide is intended for System Administrators, DevOps Engineers, and IT Managers. It covers advanced installation, user management, data security, and integration with corporate workflows.

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1. 🏗️ System Architecture

WebFlowMaster is based on a micro-services architecture:

- **Core Server**: API management, business logic, and orchestration.
- **Worker Engine**: Isolated instances for test execution (based on Playwright/Puppeteer).
- **Persistence Layer**: SQL database for metadata and Redis for message queues (BullMQ).
- **Storage Layer**: Dedicated system for storing screenshots and video traces.

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2. 👥 Governance and RBAC (Role-Based Access Control)

Access management is granular. Every user belongs to a role with predefined permissions:

Permission	Super Admin	Manager	Tester	Viewer
User Management	✅	❌	❌	❌
System Config	✅	❌	❌	❌
Project Creation	✅	✅	❌	❌
Test Creation	✅	✅	✅	❌
Test Execution	✅	✅	✅	❌
Report Viewing	✅	✅	✅	✅

Best Practice: Use the `Viewer` role for stakeholders who only need to monitor results without risking changes to the test suites.

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3. 🔐 Security and Data Protection

A. AES-256 Encryption

All credentials saved in the platform (e.g., user passwords for test logins) are encrypted at rest using the AES-256-GCM standard. The master encryption key must be managed via protected environment variables.

B. Secret and API Key Management

API Keys generated for CI/CD integration must be rotated every 90 days. WebFlowMaster supports automatic masking of secrets in execution logs to prevent accidental data leaks.

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4. ⚙️ Worker Configuration and Scalability

Workers are the computational engine. In high-load environments, you can scale horizontally by adding more Workers.

- **Concurrency**: Each worker can handle a limited number of parallel browser instances (Recommended: 2 for each CPU core).
- **Sandbox**: Each execution occurs in a clean browser context to avoid cache or cookie contamination.

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5. 🔌 Enterprise Integrations (CI/CD, Webhooks)

Pipeline Integration (GitHub/Jenkins/GitLab)

WebFlowMaster exposes a set of RESTful APIs for triggering tests. **Example of GitHub Actions integration:**

```
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - name: Trigger WebFlowMaster Test
        run: |
          curl -X POST "https://webflow-api.company.com/v1/execute" \
          -H "Authorization: Bearer ${{ secrets.WFM_TOKEN }}" \
          -d '{"planId": "prod-smoke-test"}'
```

Event-Driven Webhooks

Configure Webhooks to notify external systems (Slack, Microsoft Teams, Jira) upon critical test failure.

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6. 🛠️ Maintenance and Disaster Recovery

- **Database Backups**: Daily SQL DB backups are recommended.
- **Log Rotation**: System logs are rotated daily and kept for 30 days.
- **Health Checks**: Constantly monitor the `/api/health` endpoint to verify the status of all services (DB, Redis, Worker).

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